Function	AWS WS	Microsoft AVD
Monitoring	Cloud Watch, Cloud Trail, Systems Manager	Azure Monitor
Cost optimisation	Cost Optimiser	Advisor
Security	AWS Security Hub, Cloud Trail	Sentinel, Security Center
Serverless automation	Lambda	Azure Function
Deployment automation (IaC)	Resource Manager	Azure Resource Manager
Artifacts storage	S3 Bucket	Blobs

Table 1: Cloud native tools available with public cloud vendors

Traditional on-premises workspaces deploy isolated tools to observe these patterns. Each tool deploys data collection agents, telemetry middleware, data crunching frameworks, AI/ML algorithms, data visualisation consoles, and back-end server infrastructure setups. But most public cloud vendors provide SaaS services for many of these functions. These services simplify workspace observability framework deployment in the cloud.

Blueprint for cloud workspace observability

Figure 1 shows a typical cloud workspace observability blueprint.

Instrumentation should be built into every workspace resource and control layer to collect telemetry data. Telemetry data could be related to metrics, logs, traces, topology discovery and predictions.

Metrics that are important for workspace observability include connection attempts, session launch times, in-session latency, and desktop resource metrics (CPU, memory usage, storage performance).

Logs that are relevant for workspace observability are admin activity logs, active directory activity logs, session host logs, platform diagnostic logs, key vault logs, etc.

Traces show the activity for an individual transaction or request as it flows through an application. They are a critical part of observability as they provide context for other telemetry.

Service topology and dynamic discovery of service components are essential elements in the observability blueprint. Predictions made from historical data can also become a very useful input.

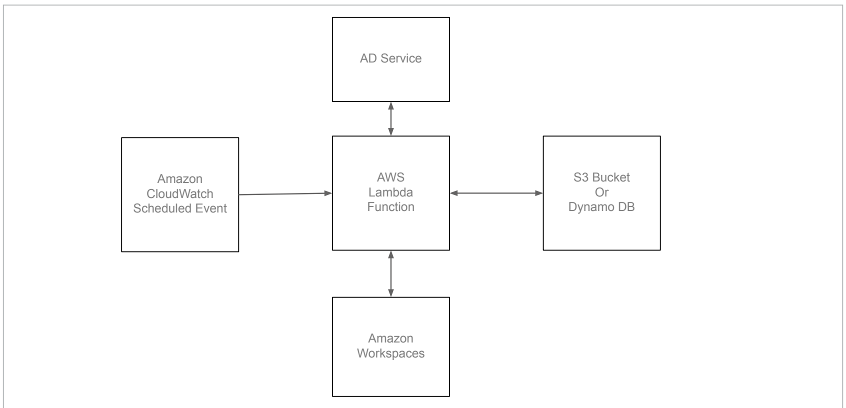

Figure 2: AWS Workspaces usage observability framework

Observability frameworks can be implemented using cloud native tools or open source tools, or a combination of the two.

Sample illustration leveraging cloud native tools

Table 1 summarises the native tools available with public cloud vendors to facilitate observability functions and patterns.

Let's now look at how to make use of cloud native services to build an observability framework.

AWS Workspaces is a managed desktop-as-a-service to provision Windows or Linux desktops. In a typical enterprise environment, some desktops are assigned to users, but they do not use them often. This leads to unnecessary costs to the company. Here is a sample solution to check the unused workspaces on a periodic basis and trigger actions to handle them. Figure 2 shows the deployment architecture.

The solution deploys Amazon Cloud Watch event rules that invoke an AWS Lambda function periodically, say for every seven days. This Lambda function checks each workspace usage. If the workspace is not used for the specified threshold duration, an action is triggered. Action could be either notifying the administrator or even de-provisioning the workspace.

To implement a comprehensive observability framework, we need to ingest and analyse logs, metrics, and traces generated by the system components and services. To handle voluminous data streams, we can make use of other cloud native services such as AWS Kinesis Firehose and data stream services.

Open source tools to complement cloud native tools

Many cloud vendors provide workspaces as a fully managed service with native monitoring tools. However, there are a few use cases like cloud vendor-agnostic monitoring, enterprise customised monitoring, multi-cloud and hybrid